# The Culture of Business, Finance and the Economy

## Class 5 – march 17

**Dr. Juan M. del Nido**

**First semester 2025**

# Today
## Mathematics enters economics:
## From reluctance to the inability to think
## wit

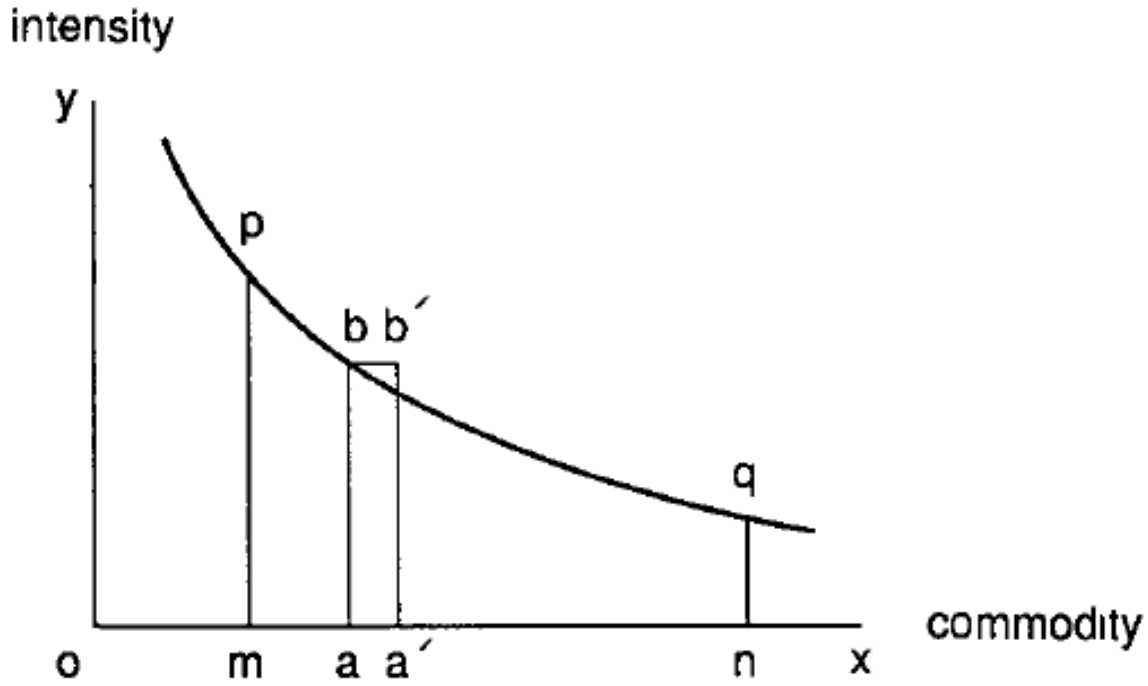



Figure 3.2 Utility Curve

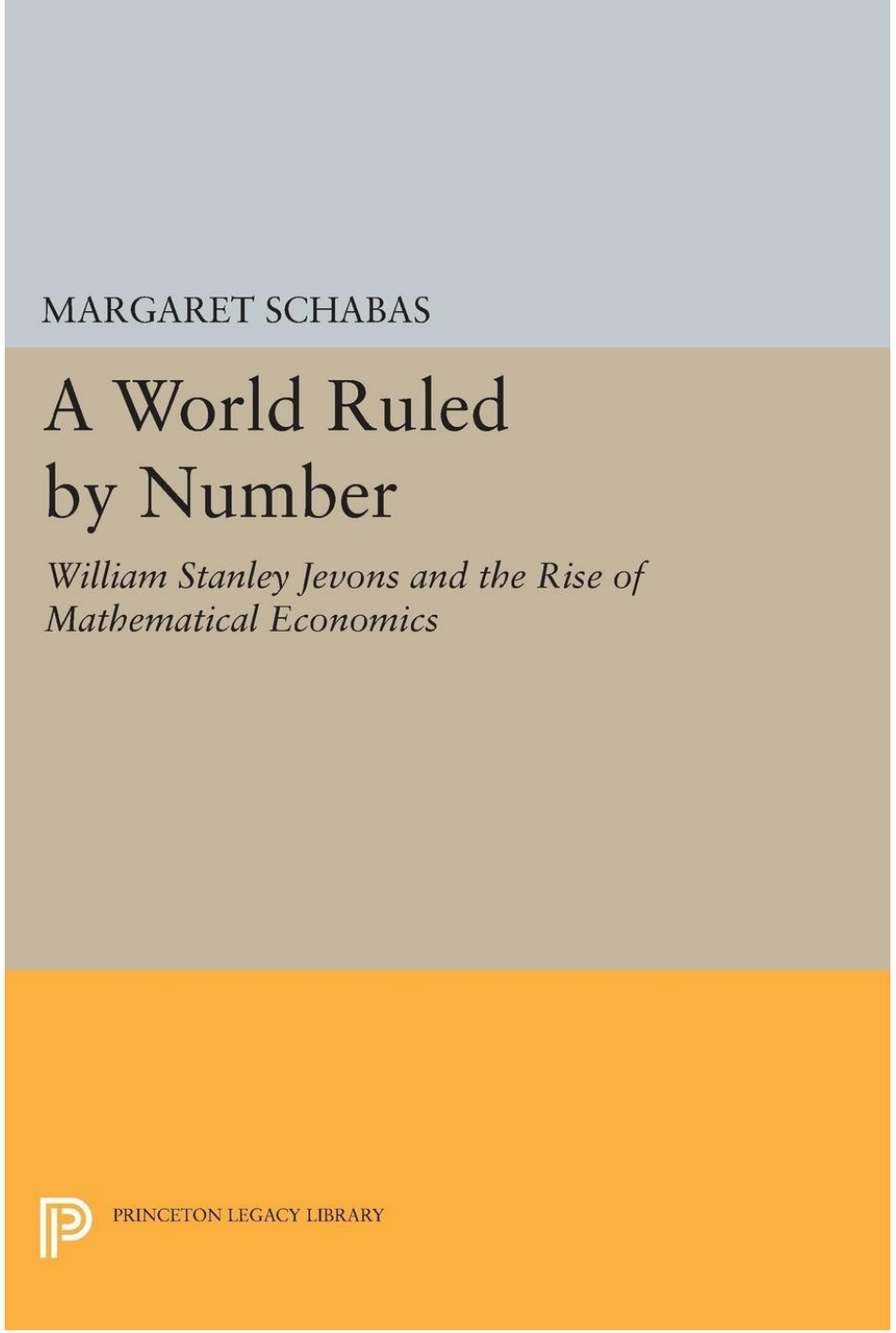
"I never learned math so I had to think"

Prof. Joan Robinson

# Recap

**“TO LEAVE NO PROBLEM UNSOLVED”**

# How economics took on mathematics



# Before 1870s: Political Economy

**Principles, not laws, specific markets (corn), industries (steelworks) or regions (Lancaster)**

**Practical knowledge**

**Non-mathematical models**

# Before 1870s

**“…the science of political economy bears a nearer resemblance to morals and politics than to mathematics.”**

**Adam Smith, Malthus, Ricardo and John Stuart Mill**

10-64

ON

# THE PRINCIPLES

OF

# POLITICAL ECONOMY,

AND

# TAXATION.

BY DAVID RICARDO, ESQ.

THIRD EDITION.

LONDON:

JOHN MURRAY, ALBEMARLE-STREET.

1821.

they were the produce of precisely the former quantity of labour. Of every hundred hats, coats, and quarters of corn produced, if

| | |
|---|---|
| The labourers had before | 25 |
| The landlords . . . | 25 |
| And the capitalists . . | 50 |
| | 100: |

And if, after these commodities were double the quantity, of every 100

| | |
|---|---|
| The labourers had only . | 22 |
| The landlords . . . . | 22 |
| And the capitalists . | 56 |
| | 100: |

In that case I should say, that wages and rent had fallen and profits risen; though, in consequence of the abundance of commodities, the quantity paid to the labourer and landlord would have increased in the proportion of 25 to 44. Wages are to be estimated by their real value, viz. by the quantity of labour and capital employed in producing them, and not by their nominal value either in coats, hats, money, or corn. Under the circumstances I have just supposed, commodities would have fallen to half their former value, and if money had not varied, to half their former price also. If then in this medium, which had not varied in value, the



and while I employed these portions, the rent would remain the same, although the produce of each should have an equal augmentation. If, instead of 100, 90, 80, 70, the produce should be increased to 125, 115, 105, 95, the rent would still be 60, or the difference between

| | | | |
|---|---|---|---|
| 95 and 125 = 30 | whilst the produce | 125 |
| 95 and 115 = 20 | would be increased | 115 |
| 95 and 105 = 10 | to 440 | 105 |
| | | 95 |
| 60 | | 440 |

But with such an increase of produce, without an increase of demand*, there could be no motive for employing so much capital on the land; one portion would be withdrawn, and consequently the last portion of capital would yield 105 instead of 95, and rent would fall to 30, or the difference between

| | | |
|---|---|---|
| 105 and 125 = 20 | whilst the produce will be still | 125 |
| 105 and 115 = 10 | adequate to the wants of the | 115 |
| 30 | population, for it would be 345 | 105 |
| | qarters, or | 345 |

* I hope I am not understood as undervaluing the importance of all sorts of improvements in agriculture to landlords—their immediate effect is to lower rent; but as they give a great stimulus to population, and at the same time enable us to cultivate poorer lands, with less labour, they are ultimately of immense advantage to landlords. A period however must elapse, during which they are positively injurious to him.

case, and only of 9.44 in the other, the rise would be as 9.44 to 10, or as 4*l.* to 4*l.* 4*s.* 8*d.* If 10 men be further employed, and the return be

| | | |
|---|---|---|
| 160, | the price will rise to | £4 10 0 |
| 150, | . . . . . | 4 16 0 |
| 140, | . . . . . | 5 2 10 |

Now if no rent was paid for the land which yielded 180 quarters, when corn was at 4*l.* per quarter, the value of 10 quarters would be paid as rent when only 170 could be procured, which, at 4*l.* 4*s.* 8*d.* would be 42*l.* 7*s.* 6*d.*

| | | | |
|---|---|---|---|
| 20 qrs. when 160 were produced, | which at £4 10 0 | would be | £90 0 0 |
| 30 qrs. | 150 | 4 16 0 | 144 0 0 |
| 40 qrs. | 140 | 5 2 10 | 205 13 4 |

Corn rent would increase in the proportion of { 100, 200, 300, 400 } and money rent in the proportion of { 100, 212, 340, 485 }



wages will be reduced; and not only his command of corn, but his general condition will be deteriorated, by his finding it more difficult to maintain the market rate of wages above their natural rate. While the price of corn rises 10 per cent., wages will always rise less than 10 per cent., but rent will always rise more; the condition of the labourer will generally decline, and that of the landlord will always be improved.

When wheat was at 4*l.* per quarter, suppose the labourer's wages to be 24*l.* per annum, or the value of six quarters of wheat, and suppose half his wages to be expended on wheat, and the other half, or 12*l.*, on other things. He would receive

| | | | | |
|---|---|---|---|---|
| 24*l.* 14*s.* | when wheat was at | 4*l.* 4*s.* 8*d.* | or the value of | 5.83 qrs. |
| 25*l.* 10*s.* | | 4*l.* 10*s.* | | 5.66 qrs. |
| 26*l.* 8*s.* | | 4*l.* 16*s.* | | 5.50 qrs. |
| 27*l.* 8*s.* 6*d* | | 5*l.* 2*s.* 10*d.* | | 5.33 qrs. |

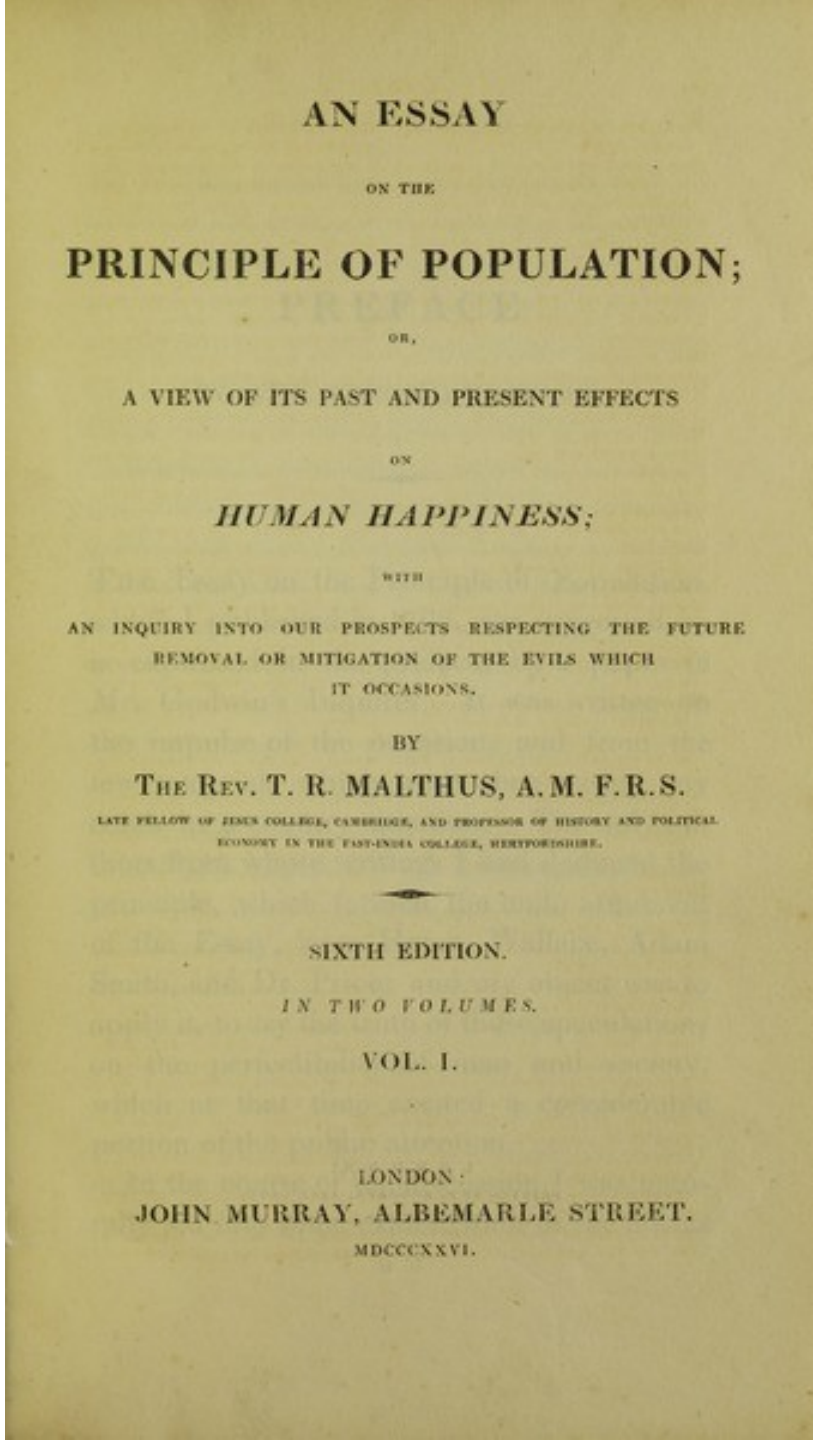
He would receive these wages to enable him to live just as well, and no better, than before; for when corn was at 4*l.* per quarter, he would expend for three quarters of corn, at 4*l.* per quarter . . . . . 12*l.*
and on other things . . . . . 12*l.*

24*l.*

When wheat was 4*l.* 4*s.* 8*d.*, three quarters, which he and his family consumed, would cost him . . . . . 12*l.* 14*s.*
other things not altered in price . . 12*l.*

24*l.* 14*s.*

# Population grows geometrically, production grows arithmetically



AN ESSAY

ON THE

PRINCIPLE OF POPULATION;

OR,

A VIEW OF ITS PAST AND PRESENT EFFECTS

ON

*HUMAN HAPPINESS;*

WITH

AN INQUIRY INTO OUR PROSPECTS RESPECTING THE FUTURE REMOVAL OR MITIGATION OF THE EVILS WHICH IT OCCASIONS.

BY

THE REV. T. R. MALTHUS, A.M. F.R.S.

LATE FELLOW OF JESUS COLLEGE, CAMBRIDGE, AND PROFESSOR OF HISTORY AND POLITICAL ECONOMY IN THE EAST-INDIA COLLEGE, HERTFORDSHIRE.

SIXTH EDITION.

*IN TWO VOLUMES.*

VOL. I.

LONDON:

JOHN MURRAY, ALBEMARLE STREET.

MDCCCXXVI.

AN

# INQUIRY

INTO THE

# Nature and Causes

OF THE

# WEALTH OF NATIONS.

By ADAM SMITH, LL.D. and F.R.S.
Formerly Professor of Moral Philosophy in the University of GLASGOW.

IN TWO VOLUMES.

VOL. I.

LONDON:
PRINTED FOR W. STRAHAN; AND T. CADELL, IN THE STRAND.
MDCCLXXVI.

# PRINCIPLES

OF

# POLITICAL ECONOMY

WITH

SOME OF THEIR APPLICATIONS TO SOCIAL PHILOSOPHY.

BY

JOHN STUART MILL.

PEOPLE'S EDITION.

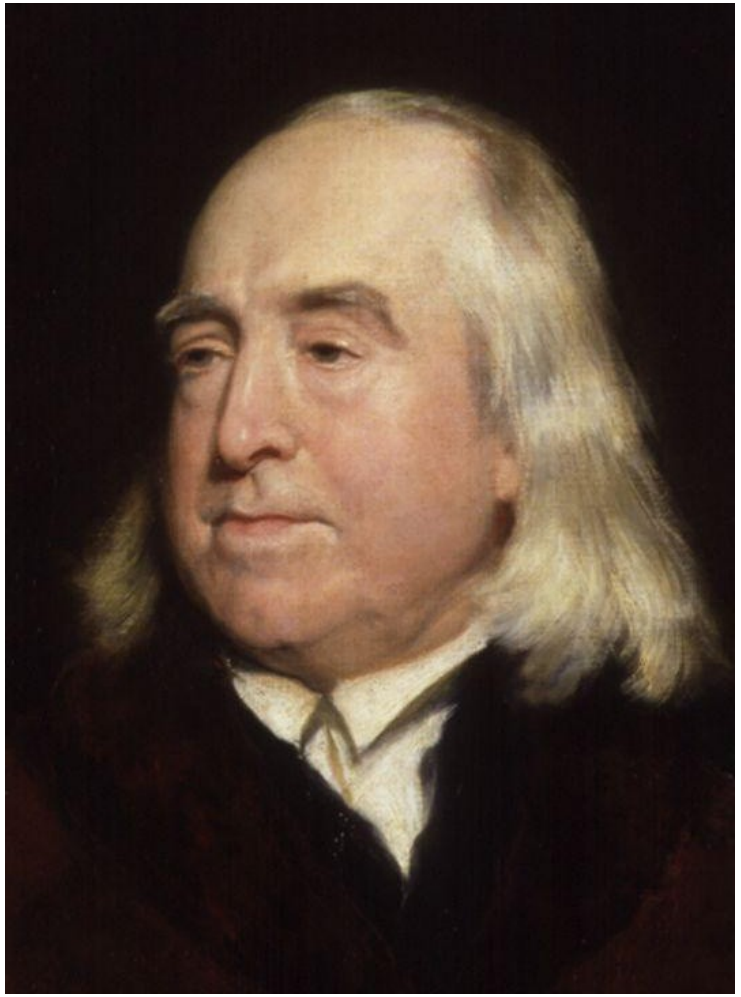
LONDON:
LONGMAN, GREEN, LONGMAN, ROBERTS, AND GREEN.
MDCCCLXV.

1865

**The Principles of Political Economy, 1848, by John Stuart Mill**

**“Happily, there is nothing in (our discipline) which remains to clear up; the theory of the subject is complete”**

# Warming up, on the benches…



**humans are creatures of pain and pleasure. They seek to *minimise* pain and *maximise* pleasure**

II. Ethics at large may be defined, the art of directing men's actions to the production of the greatest possible quantity of happiness, on the part of those whose interest is in view.

**During the 1870s**

**Labour protests, labour movements, strikes, tariff imports, growing national debt, battles about free trade…**

**… maybe not quite *everything* had been cleared up in “political economy”**

GALLANT CONDUCT OF THE POLICE.—HEROIC ATTACK UPON THE LITTLE MATCHMAKERS AND MATCHSELLERS ON THE THAMES EMBANKMENT.

# During the 1870s

# William Stanley Jevons

**“to fling aside, once and for ever, the preposterous assumptions of political economy”**

# William Stanley Jevons

**Reorganised economics on Bentham's utilitarianism: "quantities of happiness"**

**"Economics must be mathematical, simply because it deals with quantities."**

**Jevons regarded mathematics as the means to further the scientific standing of economics; mathematics was a vehicle to reach greater levels of abstraction and generality**

# William Stanley Jevons

# But… which “mathematics” to use?

# The TRIPOS

**Jevons was trying to compare how tiny amounts of goods effect tiny increases in happiness. That math was no good for that. From his travels and writings in Australia and the continent he knew of another branch of mathematics, developed by Leibniz in Germany, that could work…**

Sit ordinaria quantitas $y$, repræsentans (si placet) rectam, quanquam sub ea etiam tempus aut velocitatem aut quidvis aliud intelligere licebit. Habemus inde ascendendo infinitam, infinities infinitam, infinities infinitesies infinitam et ita porro. Rursus ab ordinaria descendendo habemus infinitesimam, infinitesimam infinitesimam, infinitesima ab infinitesima infinetisam et ita porro.

$$ddddx \mid dddx \mid ddx \mid dx \mid x \mid \int x \mid \iint x \mid \iiint x$$

ut in vulgari lege homogeneorum

$$\frac{1}{x^2} \mid \frac{1}{x} \mid 1 \mid x \mid x^2 \mid x^3 \mid x^4$$

~~Exponentes gradus tam in vulgari quam in Transcendentali lege homogeneorum sunt~~

seu

$$x^{-2} \mid x^{-1} \mid x^0 \mid x^1 \mid x^2 \mid x^3 \mid x^4$$

Et verò exponentes graduum sunt progressionis Arithmeticæ ~~[illegible]~~ in vulgari lege, ita et in nostra nova Transcendentali. Nam $x\,ddx$ et $dx^2$ sunt homogenea; quia $\int x\,dx = x\,ddx + dx^2$. ergo cum ipsig $x$ exponens sit unitas, si ipsig $dx$ exponens sit $e$, ipsig $ddx$ sit $f$, fiet $f+1 = 2e$. eodem modo $dx$ $ddx$ et $dddx^2$ sunt homogenea, adeoque exponentes Affectionum exponentes sunt progressionis Arithmeticæ. Porro quemadmodum differentia progressionis

**dy/**
**dx**

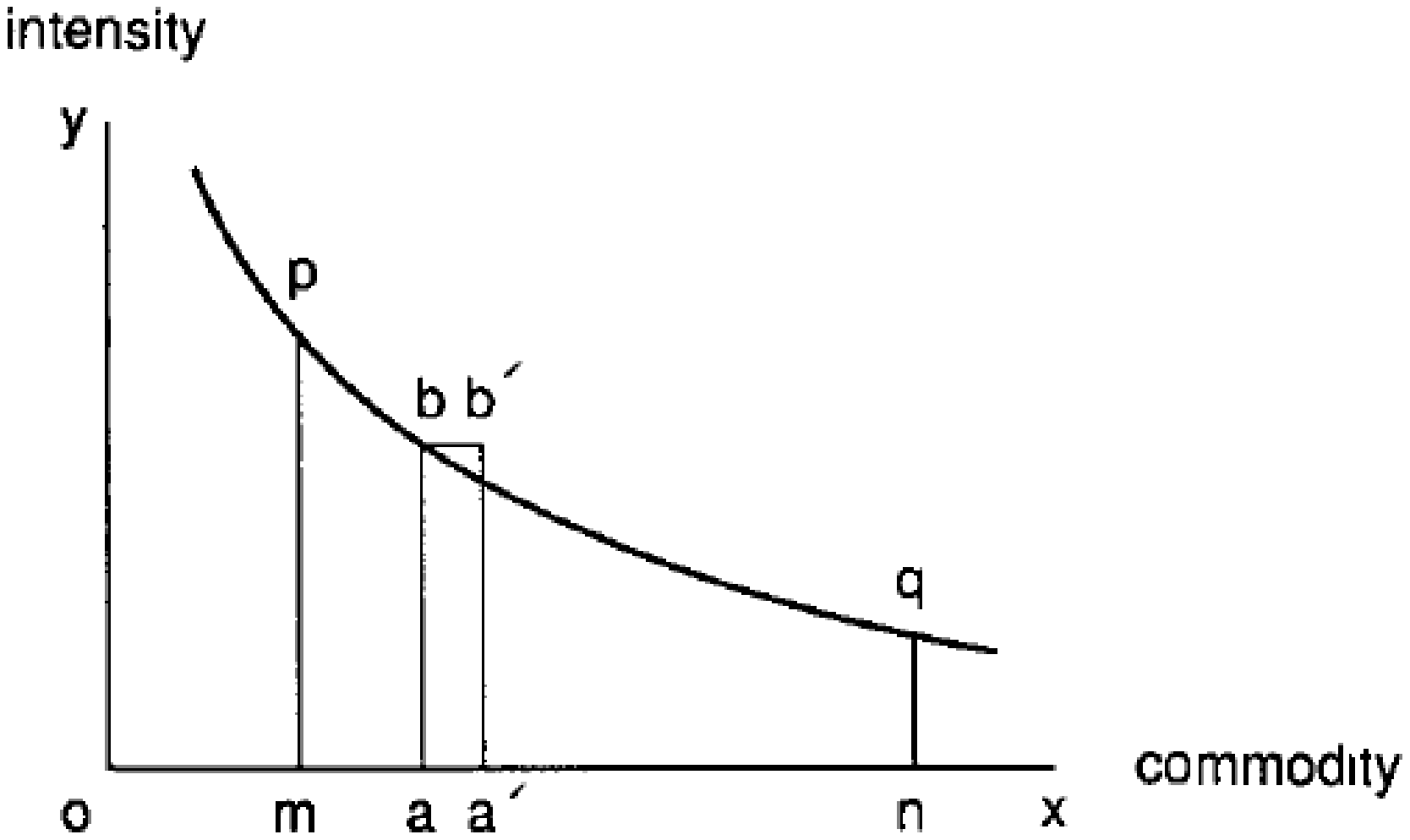



Figure 3.2 Utility Curve

# Break

# William Stanley Jevons

**With calculus he developed the principle of diminishing marginal utility: "the degree of utility varies with the quantity of commodity, and ultimately decreases as that quantity increases." Marginal utility is the bedrock of contemporary economics, and it couldn’t have existed without calculus, or in other words…**

**But calculus can’t work with “class struggle” “moral economy” or “land ownership debates”.**

**In order for economics to reach the generality Jevons though it should have, and for it to fit mathematics, all those things became “non-economic knowledge”…**

# William Stanley Jevons

**"I do not write as a mathematician, but as an economist wishing to convince other economists that their science can only be satisfactorily treated on an explicitly mathematical basis”**

**Jevons released economics from its material constraints and shifted the focus. You can now do it at your desk, no practical knowledge of farms, ships, coal, or anything needed – just**

# Reception in the field

**“I had a growing feeling in the later years of my work at the subject that a good mathematical theorem dealing with economic hypotheses was very unlikely to be good economics”**

**Alfred Marshall**

**“Though a skilled mathematician, Alfred Marshall used mathematics sparingly. He saw that excessive reliance on this instrument might lead us astray in pursuit of intellectual toys, imaginary problems not conforming to the conditions of real life: and further, might distort our sense of proportion by causing us to neglect factors that could not easily be worked up mathematically”**

**Arthur Pigou on Alfred Marshall**

# Mathematics in Economics Today

**“The economics profession went astray because economists mistook beauty, clad in impressive-looking mathematics, for truth… economists fell in love with an idealized vision of an economy gussied up with fancy equations. . . . The central cause of the profession’s failure was the desire for an all-encompassing, intellectually elegant approach that also gave economists a chance to show off their mathematical prowess”**

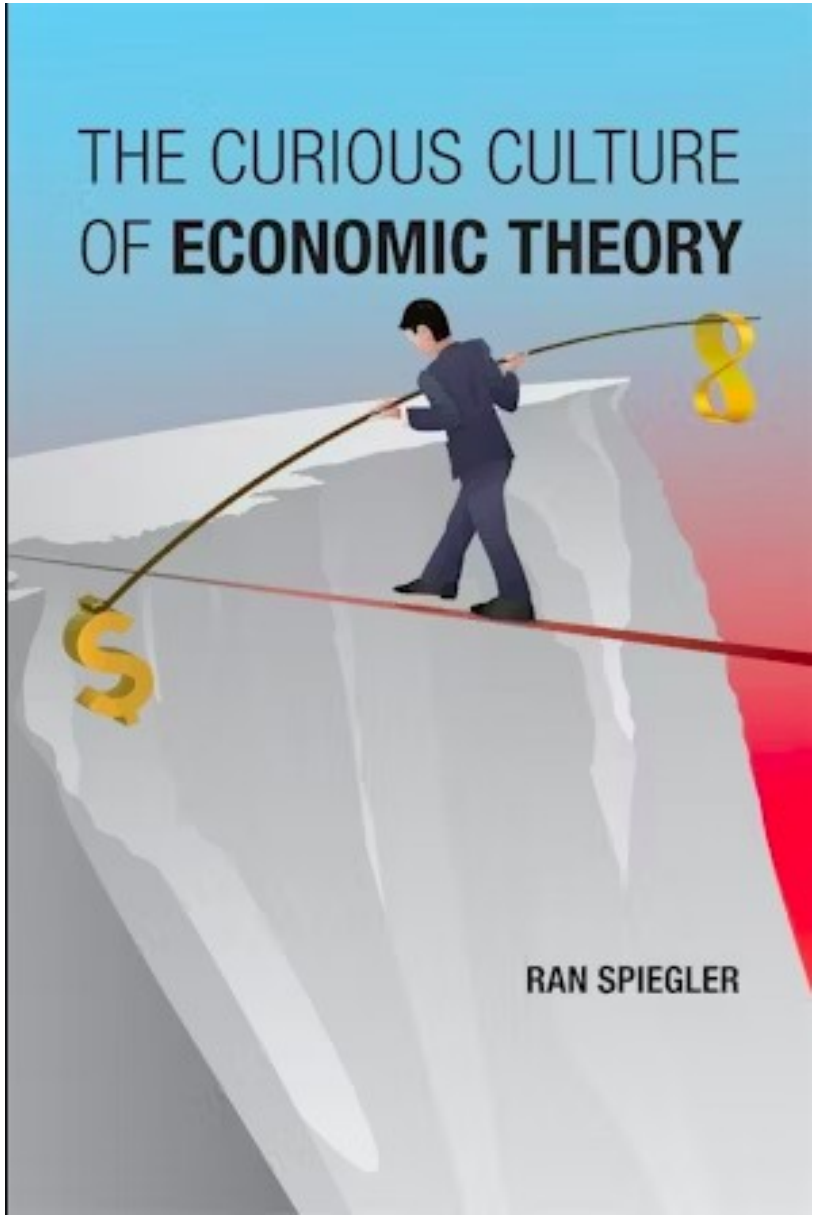
**Paul Krugman**

**“Economics has yet to get over its childish passion for mathematics and for purely theoretical and often highly ideological speculation, at the expense of historical research and collaboration with the other social sciences”**

**Thomas Piketty**

# Then… why?!?!

# 1) Rite of passage



**“(Students in Economics) are convinced that formal modelling is important to success, but are not convinced that the formal models provide deep insight into or reflect a solid understanding of the economic institutions being modeled. Believing this, they want to be trained in what the profession values. Thus we find that students who believe they are not being taught the most complicated theory feel deprived and unhappy because they worry about the ability to compete”**

**Ran Spiegler, *The Curious Culture of Economic Theory***

## 2) Status anxiety

**“Unlike engineers and chemists (or doctors, or tech wizards, etc.) economists cannot point to concrete objects – cell phones, plastic, (nuclear energy, lives saved) – to justify the high valuation of their discipline. Nor, in the case of financial economics and macroeconomics, can they point to the predictive power of their theories”**

**So mathematics makes them “feel smart” and achieving mathematical skills makes them feel valuable**

# 3) Authority in a technocratic world

**In 1986, an article in Science accused economics of misusing the language of the physical sciences to conceal ‘emptiness behind a breastwork of mathematical formulas’. Deirdre N McCloskey’s *The Rhetoric of Economics* (1998) and Robert H Nelson’s *Economics as Religion* (2001) both argued that mathematics in economic theory serves, in McCloskey’s words, primarily to say “Look at how very scientific I am.”**

# 3) The genie has left the bottle

**It is already inside the fabric of the discipline; inside its very logics. Economists have created economic phenomena through mathematics and now there's no way of knowing those phenomena other than through mathematics**

$$\overbrace{C(S,t)}^{\text{Call Price}} = \overbrace{N(d_1)}^{\text{Risk-adj. prob. that option won't expire ITM}}\underbrace{S}_{\text{Stock Price}} - \overbrace{N(d_2)}^{\text{Risk-adj. prob. that option will expire ITM}}\underbrace{K}_{\text{Strike Price}}\overbrace{e^{-rT}}^{\text{Discounts strike price to present time}}$$

# 4) It makes economic reasoning the bridge between tech, management and finance

## ¿Por qué estudiar Tecnología Digital en la Di Tella?

Nuestra carrera te ofrece:

- Una **formación multidisciplinaria única en el país**, que combina **Computación, Diseño, Neurociencia y Negocios**, para que seas parte de la creación, diseño y decisiones estratégicas del negocio, con saberes a la vez técnicos, neurocientíficos, estéticos y empresariales.

Christmas Specials | The Ivy League rat race

# Finance, consulting and tech are gobbling up top students

Are they abandoning their dreams—and does that matter?

Share

**5) a quest for truth beyond words, beyond particulars, beyond idiosyncrasies (remember Bacon)**

**Not unique to Economics, but they rule the world so these battles are fought in their terms**

# Recapping

**“There is no obvious reason why economists should ever have adopted mathematics. But one might make a similar case for Physics. It is possible to imagine a history of physics in which mathematics does not appear. The physics that we do possess, however, is fundamentally mathematical. Its central concepts of matter and force are only fully explicable in the language of mathematics. Similarly, economics could hardly have become the discipline it is at present were it not for some fruitful anchoring of mathematics at the theoretical level”**

**Margaret Schabas – *A world ruled by number***

# Conclusion

**“I never learned math so I had to think”**

**Prof. Joan Robinson**